

Q&A: Week One - Slide deck III - The normal distribution and related theorems

Version 2

Normality and linear regression

- Is normality required for ordinary least-squares point estimation?
 - No
- What does the normal-error assumption provide for linear regression?
 - Exact finite-sample distributions used for inference
- Under normal errors, how are least-squares estimates related to maximum likelihood estimates?
 - They are also maximum likelihood estimates

The univariate normal distribution

- How is a univariate normal random variable with mean μ and variance σ^2 denoted?
 - $X \sim N(\mu, \sigma^2)$
- What is the probability density function of $X \sim N(\mu, \sigma^2)$?

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}, \quad x \in \mathbb{R}.$$

- What is the mean of X ?
 - $E(X) = \mu$
- What is the variance of X ?
 - $\text{Var}(X) = \sigma^2$

The multivariate normal distribution

- How is a p -dimensional multivariate normal random vector denoted?
 - $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- What does $\boldsymbol{\mu}$ represent?
 - The mean vector of $\mathbf{X}$

- What does Σ represent?
 - The covariance matrix of $\mathbf{X}$
- What condition on Σ is required for the ordinary multivariate normal density?
 - Σ must be positive definite
- What is the density of $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \Sigma)$ when Σ is positive definite?

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}, \quad \mathbf{x} \in \mathbb{R}^p.$$

- What is $E(\mathbf{X})$?
 - $E(\mathbf{X}) = \boldsymbol{\mu}$
- What is $\text{Cov}(\mathbf{X})$?
 - $\text{Cov}(\mathbf{X}) = \Sigma$
- How can $\text{Cov}(\mathbf{X})$ be written as an expectation?
 - $\text{Cov}(\mathbf{X}) = E\{(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T\}$
- What is the (i, j) entry of Σ ?
 - $\text{Cov}(X_i, X_j)$

Moment-generating functions

- What is the moment-generating function of a random vector $\mathbf{X}$?

$$M_{\mathbf{X}}(\mathbf{t}) = E \left\{ \exp(\mathbf{t}^T \mathbf{X}) \right\}.$$

- What is $\mathbf{t}$ in the multivariate MGF?
 - A deterministic vector conformable with $\mathbf{X}$
- What property makes an existing MGF useful for identifying a distribution?
 - It uniquely determines the distribution
- What is the MGF of $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \Sigma)$?

$$M_{\mathbf{X}}(\mathbf{t}) = \exp \left\{ \mathbf{t}^T \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^T \boldsymbol{\Sigma} \mathbf{t} \right\}.$$

Linear transformations of multivariate normal random vectors

- What distributional condition is imposed on $\mathbf{X}$ in the linear-transformation theorem?
 - $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- What are the dimensions of the fixed transformation matrix $\mathbf{C}$?
 - $q \times p$

- If $\mathbf{Y} = \mathbf{CX}$, what is the distribution of $\mathbf{Y}$?

$$\mathbf{Y} = \mathbf{CX} \sim N_q(\mathbf{C}\boldsymbol{\mu}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^T).$$

- What is the mean of $\mathbf{Y} = \mathbf{CX}$?
 - $E(\mathbf{Y}) = \mathbf{C}\boldsymbol{\mu}$
- What is the covariance matrix of $\mathbf{Y} = \mathbf{CX}$?
 - $\text{Cov}(\mathbf{Y}) = \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^T$
- Must $\mathbf{C}$ have full row rank?
 - No
- What happens if $\mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^T$ is singular?
 - The resulting multivariate normal distribution is degenerate
- Do the mean and covariance formulas remain valid when the resulting distribution is degenerate?
 - Yes

Proof of the linear-transformation theorem

- How is the MGF of $\mathbf{Y} = \mathbf{CX}$ expressed in terms of the MGF of $\mathbf{X}$?

$$\begin{aligned} M_{\mathbf{Y}}(\mathbf{t}) &= E \left\{ \exp(\mathbf{t}^T \mathbf{Y}) \right\} \\ &= E \left\{ \exp(\mathbf{t}^T \mathbf{CX}) \right\} \\ &= E \left\{ \exp \left((\mathbf{C}^T \mathbf{t})^T \mathbf{X} \right) \right\} \\ &= M_{\mathbf{X}}(\mathbf{C}^T \mathbf{t}). \end{aligned}$$

- What does substitution of the MVN MGF give for $M_{\mathbf{Y}}(\mathbf{t})$?

$$M_{\mathbf{Y}}(\mathbf{t}) = \exp \left\{ \mathbf{t}^T \mathbf{C}\boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^T \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^T \mathbf{t} \right\}.$$

- Which distribution has this MGF?

- $N_q(\mathbf{C}\boldsymbol{\mu}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^T)$

- What conclusion follows?

- $\mathbf{CX} \sim N_q(\mathbf{C}\boldsymbol{\mu}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^T)$

Selection matrices

- What does a selection matrix do?
 - It extracts specified components of a vector
- Why can the linear-transformation theorem be applied to a selection matrix?
 - A selection matrix defines a fixed linear transformation

Partitions of multivariate normal random vectors

- How is the random vector partitioned in the partition theorem?

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}^{(1)} \\ \mathbf{X}^{(2)} \end{pmatrix}.$$

- What are the dimensions of $\mathbf{X}^{(1)}$ and $\mathbf{X}^{(2)}$?

- $\mathbf{X}^{(1)}$ is $q \times 1$ and $\mathbf{X}^{(2)}$ is $r \times 1$

- How are the mean vector and covariance matrix partitioned?

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}^{(1)} \\ \boldsymbol{\mu}^{(2)} \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}.$$

- What is the marginal distribution of $\mathbf{X}^{(1)}$?

- $\mathbf{X}^{(1)} \sim N_q(\boldsymbol{\mu}^{(1)}, \boldsymbol{\Sigma}_{11})$

- What is the marginal distribution of $\mathbf{X}^{(2)}$?

- $\mathbf{X}^{(2)} \sim N_r(\boldsymbol{\mu}^{(2)}, \boldsymbol{\Sigma}_{22})$

- Which selection matrix extracts $\mathbf{X}^{(1)}$?

- $\mathbf{C}_1 = (\mathbf{I}_q \quad \mathbf{0}_{q \times r})$
- What is $\mathbf{C}_1 \mathbf{X}$?
 - $\mathbf{C}_1 \mathbf{X} = \mathbf{X}^{(1)}$
- What is $\mathbf{C}_1 \boldsymbol{\mu}$?
 - $\mathbf{C}_1 \boldsymbol{\mu} = \boldsymbol{\mu}^{(1)}$
- What is $\mathbf{C}_1 \boldsymbol{\Sigma} \mathbf{C}_1^T$?
 - $\mathbf{C}_1 \boldsymbol{\Sigma} \mathbf{C}_1^T = \boldsymbol{\Sigma}_{11}$
- Which selection matrix extracts $\mathbf{X}^{(2)}$?
 - $\mathbf{C}_2 = (\mathbf{0}_{r \times q} \quad \mathbf{I}_r)$
- What does joint normality imply about the marginal distributions?
 - Each marginal distribution is normal
- Do normal marginal distributions by themselves imply joint normality?
 - No

Symmetric idempotent quadratic forms

- What does it mean for a matrix $\mathbf{A}$ to be idempotent?
 - $\mathbf{A}^2 = \mathbf{A}$
- What distributional condition is imposed on $\mathbf{Y}$ in the quadratic-form theorem?
 - $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I}_n)$
- What conditions are imposed on $\mathbf{A}$?
 - $\mathbf{A}$ is symmetric and idempotent
- What is the rank of $\mathbf{A}$ denoted by?
 - k
- What is the distribution of $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$?

$$\mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi_k^2.$$

- What determines the degrees of freedom of the chi-square distribution?
 - $\text{rank}(\mathbf{A}) = k$
- Why is symmetry of $\mathbf{A}$ required in the proof?
 - It permits orthogonal diagonalisation

Proof of the quadratic-form theorem

- What type of matrix $\mathbf{P}$ is used to diagonalise $\mathbf{A}$?
 - An orthogonal matrix
- What diagonal form is obtained for a symmetric idempotent matrix of rank k ?

$$\mathbf{P}^T \mathbf{A} \mathbf{P} = \begin{pmatrix} \mathbf{I}_k & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}.$$

- What transformed random vector is defined in the proof?
 - $\mathbf{Z} = \mathbf{P}^T \mathbf{Y}$
- What is the distribution of $\mathbf{Z}$?
 - $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I}_n)$
- Why does $\mathbf{Z}$ have covariance matrix $\mathbf{I}_n$?
 - $\mathbf{P}^T \mathbf{I}_n \mathbf{P} = \mathbf{P}^T \mathbf{P} = \mathbf{I}_n$
- How is the quadratic form expressed in terms of the components of $\mathbf{Z}$?

$$\mathbf{Y}^T \mathbf{A} \mathbf{Y} = \sum_{i=1}^k Z_i^2.$$

- What are the distributions of $Z_1, \dots, Z_n$?
 - They are independent standard normal random variables
- What is the distribution of the sum of their first k squared values?
 - $\sum_{i=1}^k Z_i^2 \sim \chi_k^2$
- What conclusion follows for the original quadratic form?

$$\circ \mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi_k^2$$